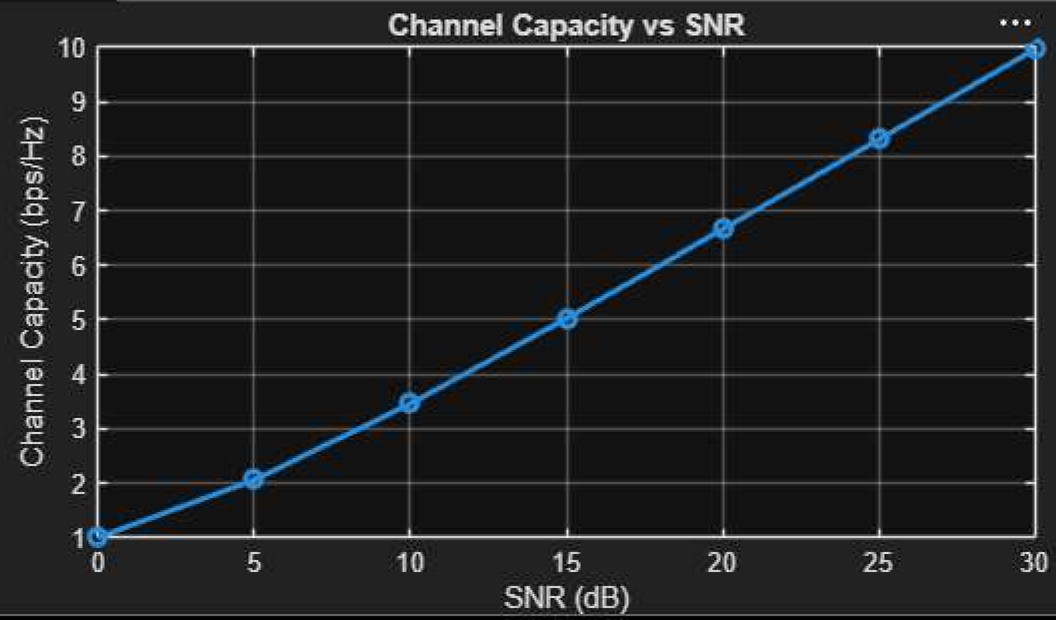
/MATLAB Drive/untitled9i.m

```matlab
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Figure 1

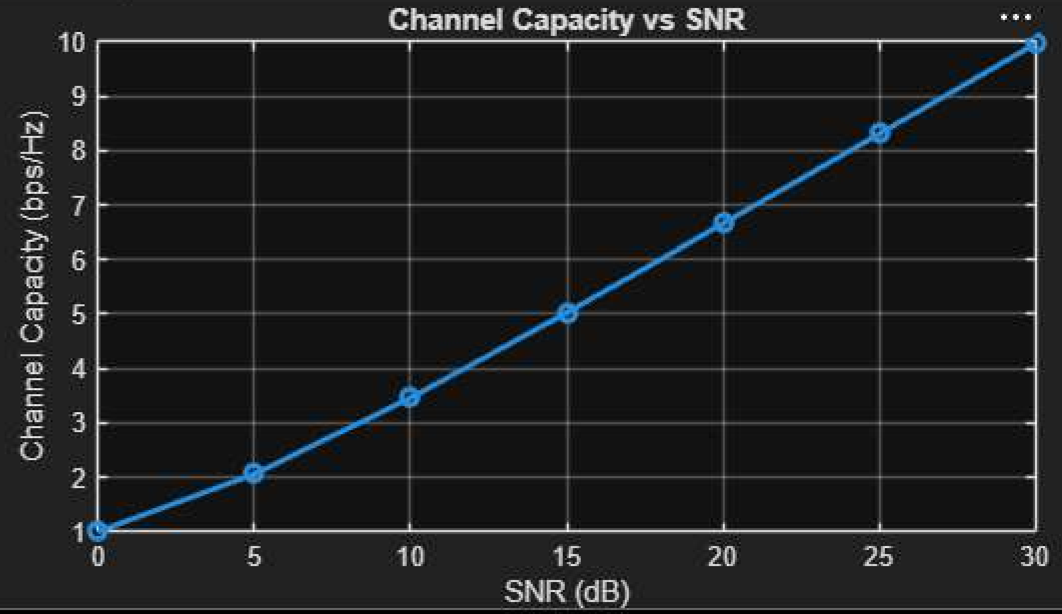


Command Window

```
>>
```

/MATLAB Drive/untitled9ii.m

```matlab
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



Command Window

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MATLAB

matlab.mathworks.com

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Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid — LABELS AND ANNOTATIONS

/ > MATLAB Drive

Files

Name
co2 at2.m
practical 1 .m
practical_2.m
practical1.m
untitled2.m
untitled2exp
untitled3.m
untitled3a5g

Workspace

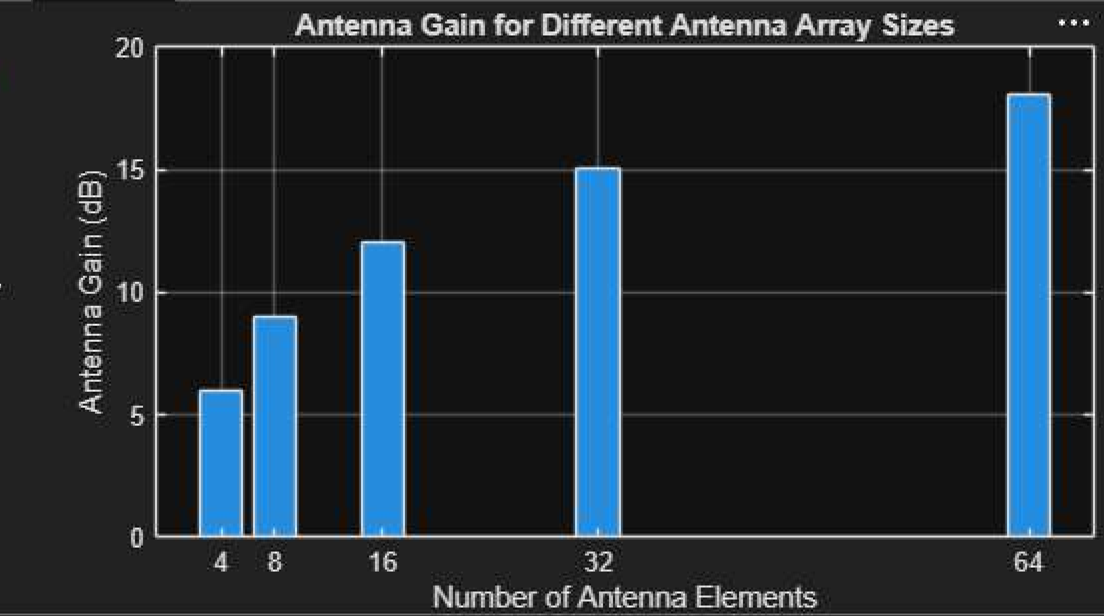
Name	Value
antennas	[4,8,16
gain	[6.020

untitled8ii.m | untitled8iii.m | untitled9i.m | untitled9ii.m | untitled9iii.m

/MATLAB Drive/untitled9iii.m

```matlab
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```

Figure 1



Command Window

```
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```

29°C Mostly cloudy | Search | ENG IN | 10:55 PM 8/11/2026